

Homework #2

(There are 8 questions here, 64 points to be scaled out of 25)

1] (8 points) Consider the following statements:

$K(x)$: "x is a Kardashian."

$L(x)$: "x is lazy."

$T(x)$: "x wastes time."

Express each of the following statements using quantifiers, logical connectives, and $K(x)$, $L(x)$, and $T(x)$. Consider the domain to be all people.

(A) "No Kardashian is lazy."

(B) "All lazy people waste time."

(C) "No Kardashian wastes time."

(D) Does (C) follow from (A) and (B) ?

2] (14 points) True or False? Consider the domain to be all real numbers. If "False", then provide a counterexample, or an explanation.

(A) $\neg \exists x(x > 1)$

(B) $\neg \exists x(x = x + 1)$

(C) $\neg \exists x(x^2 = 1)$

(D) $\neg \exists x(x + 3 = 2x)$

(E) $\forall x(x^2 \neq x)$

(F) $\forall x(|x| > 0)$

(G) $\forall x(x^2 \neq 2)$

3] (15 points)

- i) Express each of the following statements using quantifiers.
- ii) Form the negation of the statement, so that no negation is to the left of a quantifier.
- iii) Next, express the negation in simple English.

(Do not simply use the phrase "It is not the case that...")

EXAMPLE: Every horse can count.

reword this as "For every x that is a horse, x can count." Or,

"For every animal, if x is a horse, then x can count." Either (or another version) will work.

Then introduce some propositional functions:

$H(x)$: " x is a horse."

$C(x)$: " x can count."

i) $\forall x(H(x) \rightarrow C(x))$

ii) $\exists x(H(x) \wedge \neg C(x))$

iii) "There exists a horse that cannot count."

(A) Some old dogs can learn new tricks.

(B) No rabbit knows calculus.

(C) Every bird can fly.

(D) There is no dog that can talk.

(E) There is no one in this class that knows French and Russian.

4] (5 points) Determine whether the following compound proposition is satisfiable:

$(p \vee q \vee \neg r) \wedge (p \vee \neg q \vee \neg s) \wedge (p \vee \neg r \vee \neg s) \wedge (\neg p \vee \neg q \vee \neg s) \wedge (p \vee q \vee \neg s)$

5] (4 points) Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent.

6] (4 points) Find the negation of the following statements:

(A) Barnaby will attend CU or CSU.

(B) Samson is strong and fast.

7] (5 points) Three gentlemen living in the fraternity house tau tau tau decide to go to Half-fast for a sandwich. Penelope is working at the cash register there. She asks them: "Does everyone want a drink as well?"

Tod says "I do not know."

Tad says "I do not know."

Ted says "No, not everyone wants a drink."

Penelope then gives drinks to those who want them.

To whom did Penelope give drinks? Explain how you know.

8] (9 points) Elon Musk finally succeeds in creating a vehicle that can travel to space. He is so successful that he actually lands on a previously unknown planet. The planet consists of three types of creatures:

Truffys - They always tell the truth.

Flossys - They always lie.

Ambigs - Sometimes they tell the truth, sometimes they lie.

Upon landing on the planet, Elon is greeted by three of the planet's inhabitants; actually one of each type. Although the three inhabitants know what type each other are, Elon does not.

Consider three different scenarios:

(Scenario 1)

Inhabitant 1 says "I am not an Ambig."

Inhabitant 2 says "I am not an Ambig."

Inhabitant 3 says "I am not an Ambig."

(Scenario 2)

Inhabitant 1 says "I am a Truffy."

Inhabitant 2 says "I am a Truffy."

Inhabitant 3 says "I am a Truffy."

(Scenario 3)

Inhabitant 1 says "I am not an Ambig."

Inhabitant 2 says "I am not an Ambig."

Inhabitant 3 says "Inhabitant 1 is an Ambig."

For each scenario determine whether there is a unique solution and determine who the Truffy, Flossy, and Ambig are. If there is more than one solution, list it. If there is no solution, state why.